



ST BENEDICT'S

SUBJECT

**Physical Science
Paper 1**

DATE

22 July 2019

GRADE

11

MARKS

165

**EXAMINER
NAME**

Mr McCoy

MODERATOR

Mr Ntombela

CLASS

DURATION

180 minutes

**LOW
ORDER**

50%

HIGH ORDER

50%

QUESTION 1- MULTIPLE CHOICE- MR S. RAMSURWAJ

- 1.1 D✓✓
- 1.2 D✓✓
- 1.3 A✓✓
- 1.4 A✓✓
- 1.5 B✓✓
- 1.6 B✓✓
- 1.7 A✓✓
- 1.8 D✓✓

QUESTION 2 – MR Z. NTOMBELA

2.1.1 the rate of change of position✓✓ or the rate of displacement or the rate of change of displacement and know that velocity is a vector quantity (2)

2.1.2. $30 \text{ m.s}^{-1} \times 3.6 = 108 \text{ km.h}^{-1}$ ✓
Therefore the speed limit was broken. (3)

2.1.3 $v = u + at$ accept gradient calculation
 $24 = 20 + a12$ ✓
 $a = 0,33 \text{ m.s}^{-2}$ ✓ (3)

2.1.4 Car Q = $\frac{1}{2} b.h$ ✓ (formula)
 $= 0,5 \times 12 \times 30$
 $= 180 \text{ m}$ ✓

Car P = $\frac{1}{2} b.h + l \times b$
 $= 0,5 \times 12 \times 4 + 20 \times 12$
 $= 24 + 240$
 $= 264 \text{ m}$ ✓

distance between the cars = $264 - 180 = 84 \text{ m}$ ✓ (4)

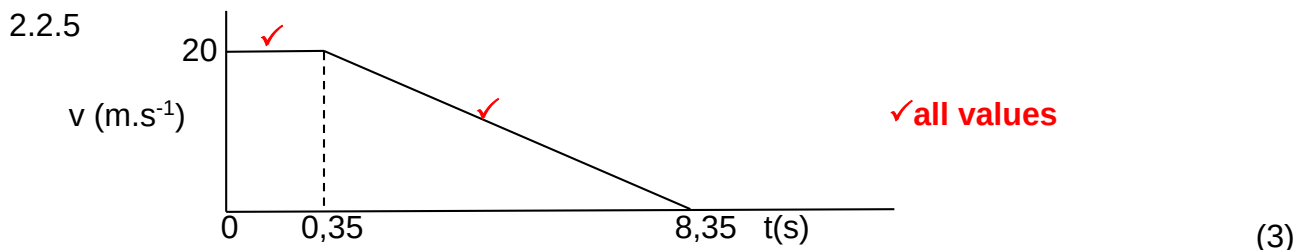
2.1.5 $v = s/t$
 $180 + 30t = 264 + 24t$ ✓✓
 $30t - 24t = 264 - 180$
 $t = 14 \text{ s}$ ✓
Time = $14 + 12 = 26 \text{ s}$ ✓ (4)

2.2.1 $0,35 \text{ s}$ ✓ (1)

2.2.2 $s = v \times t$
 $= 20 \times 0,35$
 $= 7 \text{ m}$ ✓✓ (2)

2.2.3 $v = u + at$
 $0 = 20 + (-2,5)t$ ✓ Must show that the acceleration is negative.
 $t = 8 \text{ s}$ ✓
Total time to stop = $8 + 0,35 = 8,35 \text{ s}$ ✓ (4)

2.2.4 $s = \frac{v+u}{2} t$ ✓
 $= \frac{0+20}{2} 8$ ✓
 $= 80 \text{ m (YZ)}$ ✓
car from X → $7 + 80 + 2 = 89 \text{ m}$ ✓ (4)



2.3 $s = ut + \frac{1}{2} at^2$
 $R = 0 + \frac{1}{2} at^2$ ✓
 $2R = at^2$
 $s = 4R$ ✓
 $s = 0 + \frac{1}{2} a(2t)^2$ ✓
 $s/2 = at^2$ (✓link) (4)

QUESTION 3 – MR D. DU TOIT

3.1.1 the rate of change of velocity✓✓ (2)

3.1.2 uniform acceleration✓ down✓
decreasing velocity✓ up✓ (4)

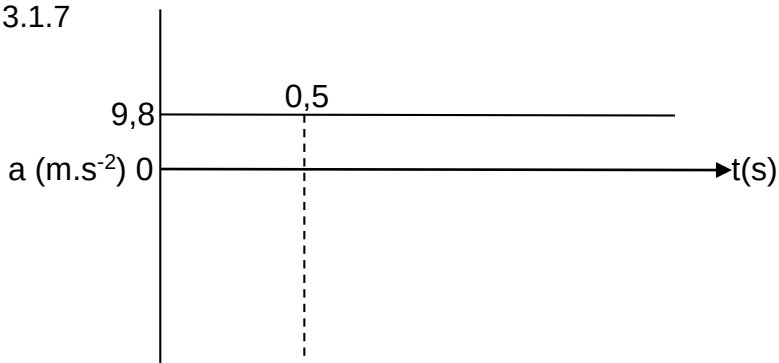
3.1.3 $s = (u+v)/2 \times t$ OR $A = \frac{1}{2} b h$
 $= (0+3,2)/2 \times 0,5$ $= \frac{1}{2} 3,2 \times 0,5$
 $= 0,8 \text{ m}$

3.1.4 dot✓ and weight✓ ONLY (2)

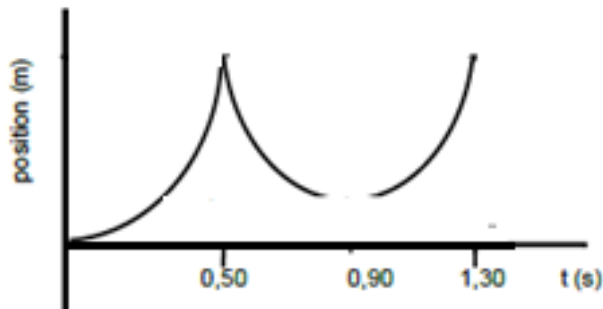
3.1.5 Yes, the velocity before and after ✓ the bounce is the different✓. (2)

3.1.6 $2,56 \text{ m.s}^{-1}$ ✓✓ (2)

3.1.7 (3)



3.1.8 ✓ - draw as shown, $\frac{3}{4}$ if on the negative side.



(4)

3.2.1 speed = distance/time
 $= 400✓ / 105 ✓$
 $= 3,81 \text{ m.s}^{-1}✓$ (3)

3.2.3 c.o.e. $3,81 \text{ m.s}^{-1}✓$ west ✓ (2)

3.2.3 distance as the length of path travelled✓,
 if the runner ran from point A to B to C✓
 displacement as a change in position✓,
 which the distance from A straight to C in a North direction. ✓ (4)

QUESTION 4 – MR D. DU TOIT

4.1.1 the single **force** which has the same effect✓ as the original forces acting together ✓ (2)

4.1.2 $x = 6000\sin 18/6000\cos 72 - \checkmark 4500\sin 50/4500\cos 40$
 $x = 1854,10 - 3447,2$
 $= -1593,01\checkmark \dots C$

$y = 6000\cos 18/6000\sin 72 + 4500\cos 50/4500\sin 40$
 $y = 5706,34 + 2892,54$
 $= 8598,88\checkmark \dots F$

$$a^2 + b^2 = c^2$$

$$-1593,01^2 + 8598,88^2 = c^2$$

$$c = 8745,21 \text{ N } \checkmark$$

$$\theta = \tan^{-1} \frac{8598,88}{1593,01}$$

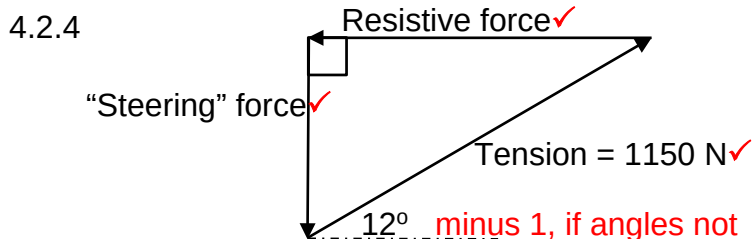
$$\theta = 79,50\checkmark$$

The resultant force on the tugboat will be 8745,21 N on a bearing 190,5 degrees. ✓ (6)

4.2.1 the rate of change of distance✓✓(AON) (2)

4.2.2 Component of the tension = $1150 \text{ N} \times \cos 12^\circ \checkmark = 1125 \text{ N} \checkmark$
 Therefore resistive forces = $1125 \text{ N} \checkmark$ (3)

4.2.3 As the tension will have a component ✓ pulling the canal boat off course, so the boat will have to be steered in the opposite direction✓ to keep it in the direction of motion. (2)



(3)

4.2.5 Longer rope the smaller the angle will become✓
 So the same component will be achieved with a lower tension. ✓ (2)

QUESTION 5 – MR F. MCCOY

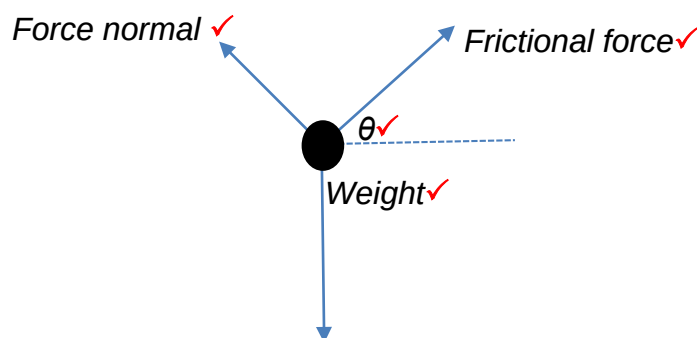
5.1 An object continues in a state of rest or uniform✓ (moving with constant) velocity unless it is acted upon by a net or resultant force✓ (2)

5.2 *the force that opposes the motion of an object* ✓ and acts parallel to the surface with which the object is in contact✓ (2)

5.3.1 The external force is applied to the truck✓
The inertia ✓ of the box resists the change in motion.
The friction on the box will not be great enough✓ to accelerate the box with the truck, ✓
hence it will slide off. (4)

5.3.2 Less than, and once the box is in motion the surfaces interact less. (3)

5.4 (4)



5.5. The box pulling up✓ on the Earth. ✓ (2)

5.6 $F_N = - \text{✓} F_g(mg) \cos \theta \text{ ✓}$ (2)

$$5.7 \quad F_f^{max} = \mu F_N \text{ ✓}$$

$$\mu = \frac{mg \sin \theta \text{ ✓}}{mg \cos \theta \text{ ✓}}$$

$$= \tan \theta$$

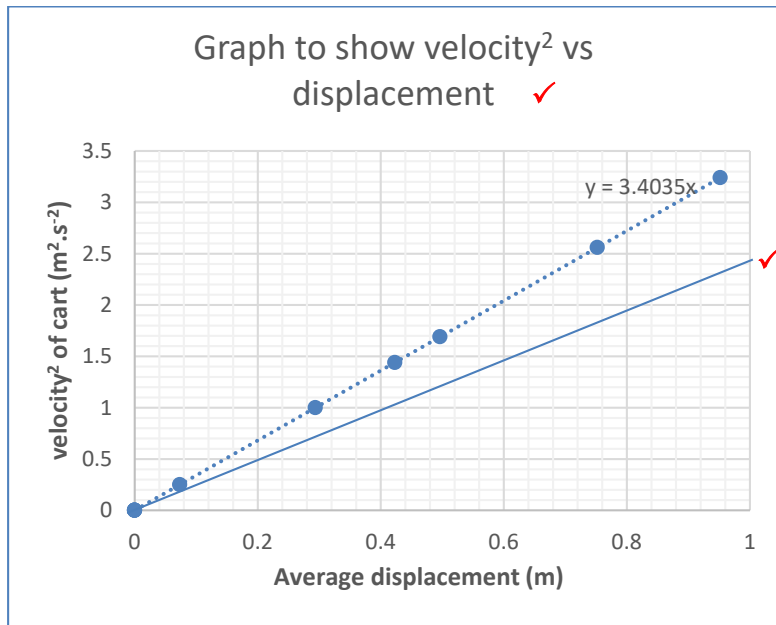
$$0,32 = \tan \theta \text{ ✓} \quad \text{MUST show how you got tan}$$

$$\theta = 17,74^\circ \text{ ✓}$$

(5)

QUESTION 6 – MR S. RAMSURWAJ

- 6.1 As the initial speed of the truck increases ✓ so the length of the ramp will ✓ have to increase. ✓ **order matters** (3)
- 6.2 By pushing it with different forces. ✓✓ (2)
- 6.3 3,24 ✓ (1)
- 6.4 & 6.5 (6 & 2)



✓ heading

✓ labels, correct unit

✓ scale

✓✓ plotting

✓ LOBF

✓ 6.5

QUESTION 7– MR F. MCCOY

7.1 the total energy supplied per coulomb of charge by the cell. ✓✓ (2)

7.2. $R_s = 6 + 12$
 $= 18 \Omega$ ✓✓ (2)

7.3 A low resistance connection ✓ (established by accident or intention.) (1)

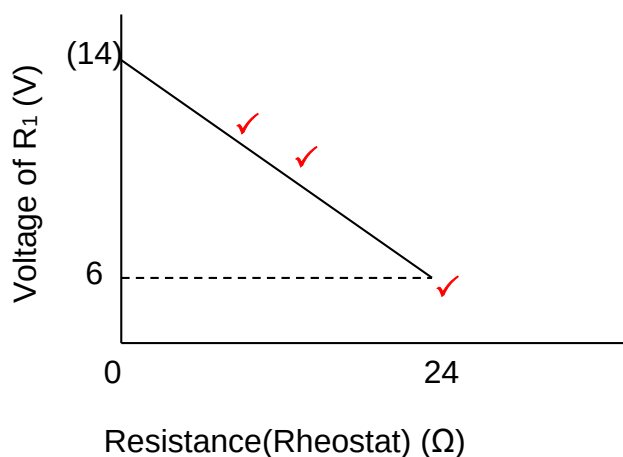
7.4.1 $I = V/R$ ✓
 $= 14 / 6$ ✓✓ if the rheostat is at zero, not current will flow through R_2
 $= 2,33 \text{ A}$ ✓ (4)

7.4.2 total resistance of parallel resistors $= 8,0 \Omega$ ✓✓
current $= 14 / (6.0 + 8.0)$
 $= 1 \text{ A}$ ✓✓ (4)

7.5.1 $P = EI$ (allow $P = VI$) or $P = V^2 / R$ or $P = I^2 R$
power $= (14 \times 1)$ ✓ or $(14^2 / 14)$ or $(1^2 \times 14)$ c.o.e.
 $= 14 \text{ W}$ ✓ (2)

7.5.2 $V = I \times R$ ✓
 $= 1 \times 6$ ✓
 6 V ✓ (3)

7.4 (3)



7.5.1 decrease ✓ (1)

7.5.2 As circuit R increases I decreases ✓ $I = V/R$ ✓
As I decreases P decreases since $P = I^2 R$ ✓ and R is constant ✓ (4)

END OF EXAMINATION